How Unified is Your Encoder?
Measuring Representational Convergence in Any-to-Any Vision Models

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CS-503 Visual Intelligence — Project Progress Report 1

I. Milestone Progress

A. Problem and milestone overview

Any-to-any models such as 4M [1] share Transformer encoder weights across modalities, but route each modality through its own learned embedding. Whether the encoder *unifies* these inputs into a shared representation, or behaves as an ensemble of disjoint modality-specific sub-networks [2], is open. RQ1 of our proposal addresses this with a multi-metric protocol (CKA [3], PWCCA [4], k-NN overlap) layered over a mismatched-scene null distribution, so that observed similarity is read against an empirical noise trajectory rather than in absolute terms.

The first milestone was scoped around *building a strong, reusable benchmarking pipeline*, not around breadth of modalities. Concretely, we restricted ourselves to a Hypersim [5] subset and a single modality pair (RGB↔Depth), and instead invested effort into the three-metric stack and the null-hypothesis machinery so that any future pair plugs in without re-engineering. This deliberately trades the breadth promised by the proposal (RGB, depth, normals, segmentation) for confidence that the protocol itself is correct before scaling. Extending to additional modalities and to DIODE [6] is the explicit goal of milestone 2; RQ2's exploratory protocols (cross-modal transfer, residual analysis) remain in the W5–W6 slot of the proposal timeline.

B. Pipeline

The benchmark proceeds in four steps.
(i) Each modality of each scene is forwarded through 4M independently; forward hooks capture the output token matrix at each of the 12 encoder blocks, yielding per-layer activation matrices indexed by scene, modality, and layer.
(ii) For every layer and every modality pair, we compute CKA (global geometric alignment), PWCCA (directional alignment of principal axes), and k-NN overlap (local topological similarity); the three target complementary facets of unification, so cross-metric agreement sharpens any claim.
(iii) Raw similarity is confounded by shared positional encodings, tokenizer priors, and inter-scene statistical We construct a layer-wise empirical null by recomputing each metric on 1,000 mismatched cross-scene modality pairs, yielding a noise trajectory across depth. To prevent the null from being inflated by scenes that share the same semantic content, left and right scenes are constrained to have *different scene types* (e.g. bathroom vs. office) drawn from the Hypersim trajectory metadata; this ensures the null reflects genuine cross-content misalignment rather than intra-category similarity.
(iv) Matched values are tested against this null per layer: we report p-values, null mean and standard deviation, and the matched−null gap.
The four steps are decoupled —the comparison step reruns without re-extracting activations— and all artefacts are logged to W&B.

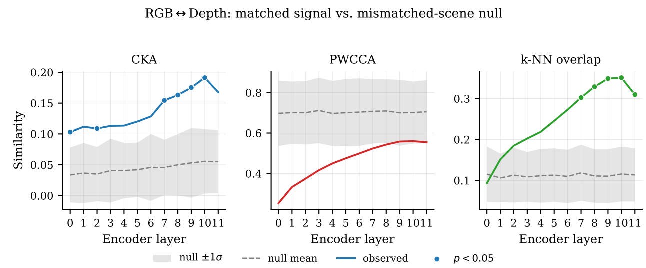

Figure 1. Layer-wise convergence on Hypersim (RGB↔Depth, n=4325, 1000 null draws).

C. Experiment

We ran the pipeline end-to-end on a Hypersim subset, 4325 aligned RGB↔Depth samples, with 1000 mismatched null draws across the 12 encoder blocks of 4M-7B-CC12M. All three metrics produced enriched outputs and the corresponding plots.

D. Results

Two of the three metrics agree on a clear depth-wise pattern (Fig. 1, Fig. 2). **CKA** rises monotonically from 0.10 at layer 0 to 0.19 at layer 10 (z=2.60, p=0.028), then dips to 0.17 at the final block. **k-NN overlap** shows the strongest signal: 0.09 at layer 0 grows to 0.35 at layer 9 (z=3.62, $p \approx 10^{-3}$) and remains highly significant through layer 11. Both shapes match the "persistent partial convergence" outcome anticipated by Fig. 2 of our proposal: significance grows with depth and the mild retreat at the last block is consistent with task/decoder specialisation. Crucially, CKA peaks at 0.19, far from 1, supporting the proposal's working

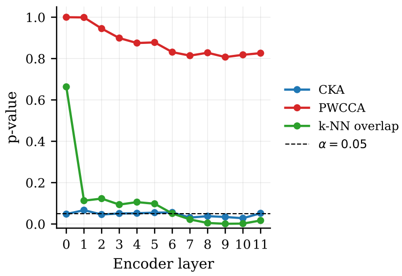

Figure 2. Per-layer p-values for the three metrics. k-NN overlap drops well below threshold from layer 7 onward; CKA hovers around it; PWCCA never reaches significance.

definition of *partial* unification rather than a fully shared trunk: the encoder pulls RGB and depth toward a common subspace without collapsing them onto it.

PWCCA disagrees. Observed values (0.25–0.56) lie below the null mean (≈ 0.70) at every layer; all p-values exceed 0.8, z-scores are negative. We interpret this as a metric pathology in our setup; one plausible explanation is that PWCCA's CCA backbone recovers spurious shared subspaces driven by tokenizer and positional artefacts that are shared across scenes regardless of content, inflating the null relative to truly matched pairs. Confirming this diagnosis is a milestone-2 priority.

E. Deviations from the proposal

Two deliberate scope changes versus the proposal: (i) the milestone-1 pipeline run uses one modality pair instead of all four, on a Hypersim subset; the breadth is deferred to milestone 2 once the protocol is trusted. (ii) PWCCA is uninformative under our current null construction, so its weight in the final claim will likely shift onto CKA and k-NN overlap unless we recover it via a stricter null. The research question, datasets, and overall plan are unchanged.

II. Discussion & Next Steps

Reading: Across two of three metrics the encoder shows depth-correlated, statistically credible alignment between RGB and depth that peaks at layers 9–10 and retreats at layer 11. In the language of the proposal's working definition, this is evidence for *partial* unification with a specialised tail—neither the "red" fragmented outcome of Fig. 2 nor a fully unified encoder.

What worked: The pipeline is functional, optimised end-to-end, and lightweight enough to be fully replicable: a complete run with 1000 null draws finishes in minutes on a single GPU, and re-running only the significance step takes seconds.

What did not work, and why: PWCCA never reaches significance, and observed values sit *below* the null mean at every layer. We do not yet have a confident explanation; diagnosing whether this stems from the metric, the null construction, or the data is on the milestone-2 list.

Risks: (i) *Hypersim is synthetic*; convergence on real images may differ, motivating the DIODE replication. (ii) *One pair only* cannot distinguish "RGB and depth converge" from "everything converges"; resolved by adding modalities. (iii) *Scale*: the current run uses a small Hypersim subset, so signal stability and generalisation must be verified on a substantially larger sample before any conclusion is treated as final.

Action items until the next report:

1) Scale the Hypersim run up to a much larger subset to verify the depth trend and tighten the null estimates.
2) Replicate on DIODE (real-world RGB+depth) using the existing adapter, isolating the synthetic-data risk.
3) Add surface normals and segmentation as additional modalities, yielding more pairs and a transitivity check on convergence.
4) Diagnose PWCCA: identify why it underperforms the null and decide whether to keep, replace, or drop it.
5) Begin RQ2 explorations (cross-modal k-NN, residual correlation) on the W5–W6 slot of the proposal timeline.

III. Author Contribution Statement

Arthur led the extraction stage and the 4M model wrapper. *Albert* and *Martina* implemented the three metrics (CKA, PWCCA, k-NN overlap) and the null-hypothesis stage. *Adrien* contributed to result analysis and the write-up. All authors participated in shaping the research question.

References

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